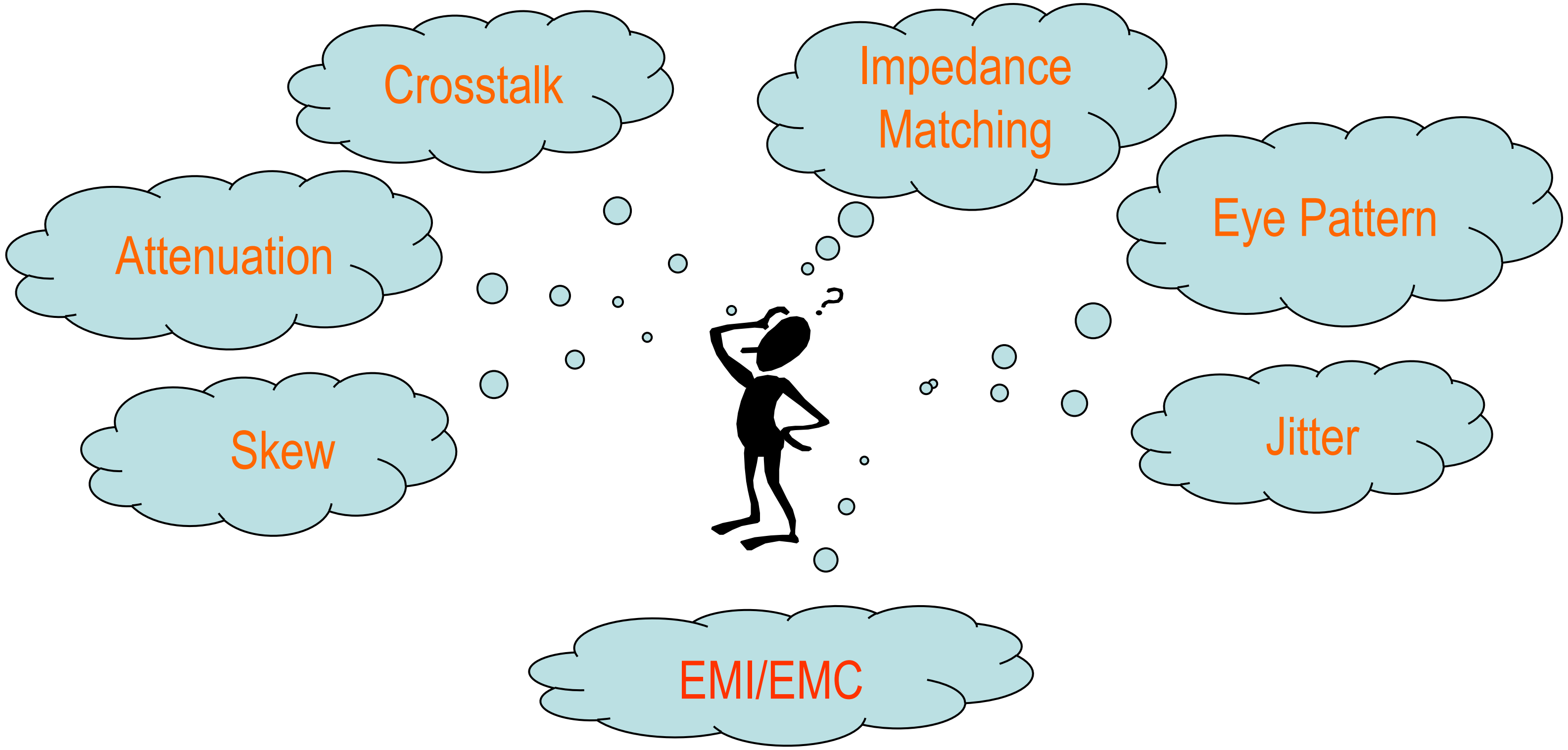
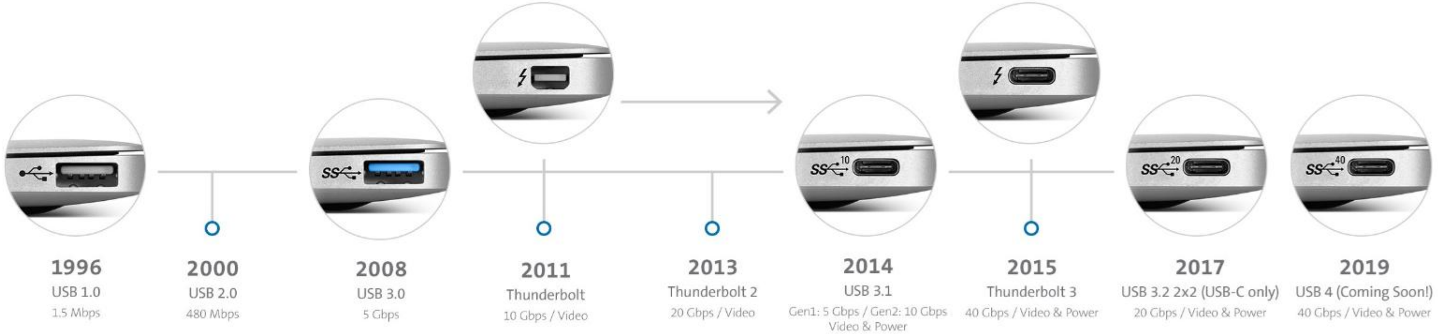
- EM/SI Simulation
- EM/SI Testing & Failure Analysis
- DAC, ACC, AEC, AOC Development
- Project Management
- Industry Standards Committee Involvement
- EM Design Automation

Challenges of High-Speed Interconnect Product Developments

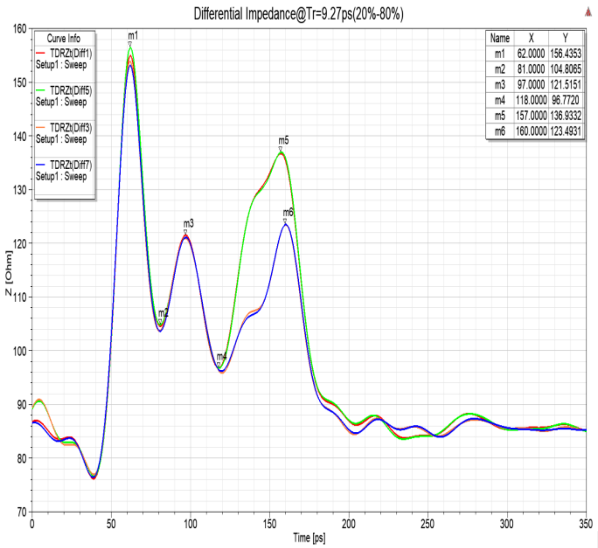


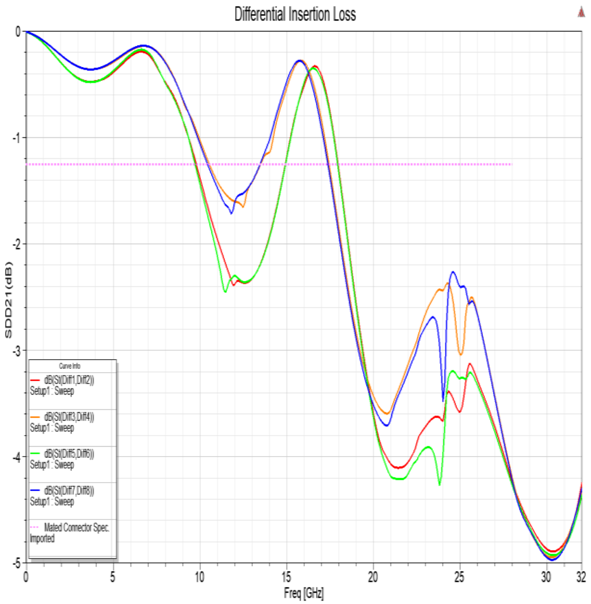


https://www.kensington.com/news/docking-connectivity-blog/the-history-of-usb-usb4-explained/

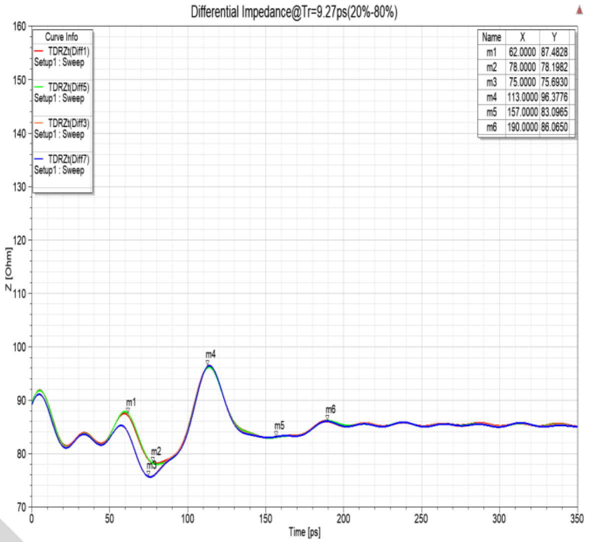
- Time-Domain: Z Match & XT Isolation
- Frequency-Domain: Resonant Freq. Migration, Common-Mode Suppression
- EMC: Shielding Design
- Circuit & Channel Simulation: Eye Diagram, Jitter, COM

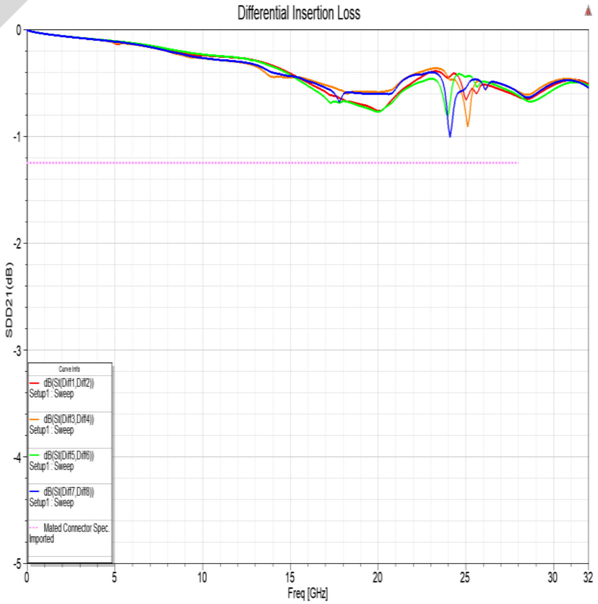
Original



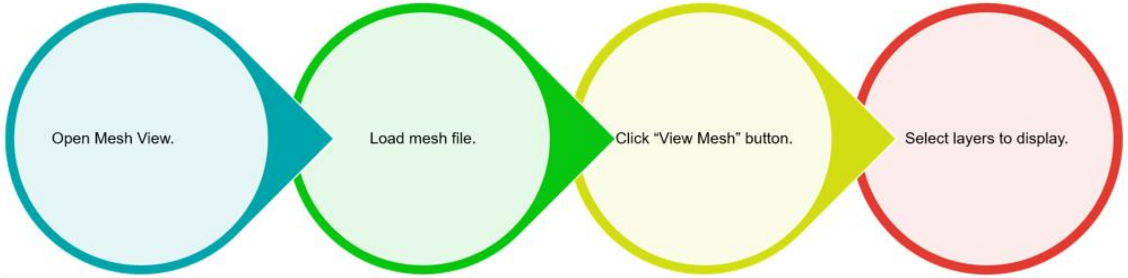


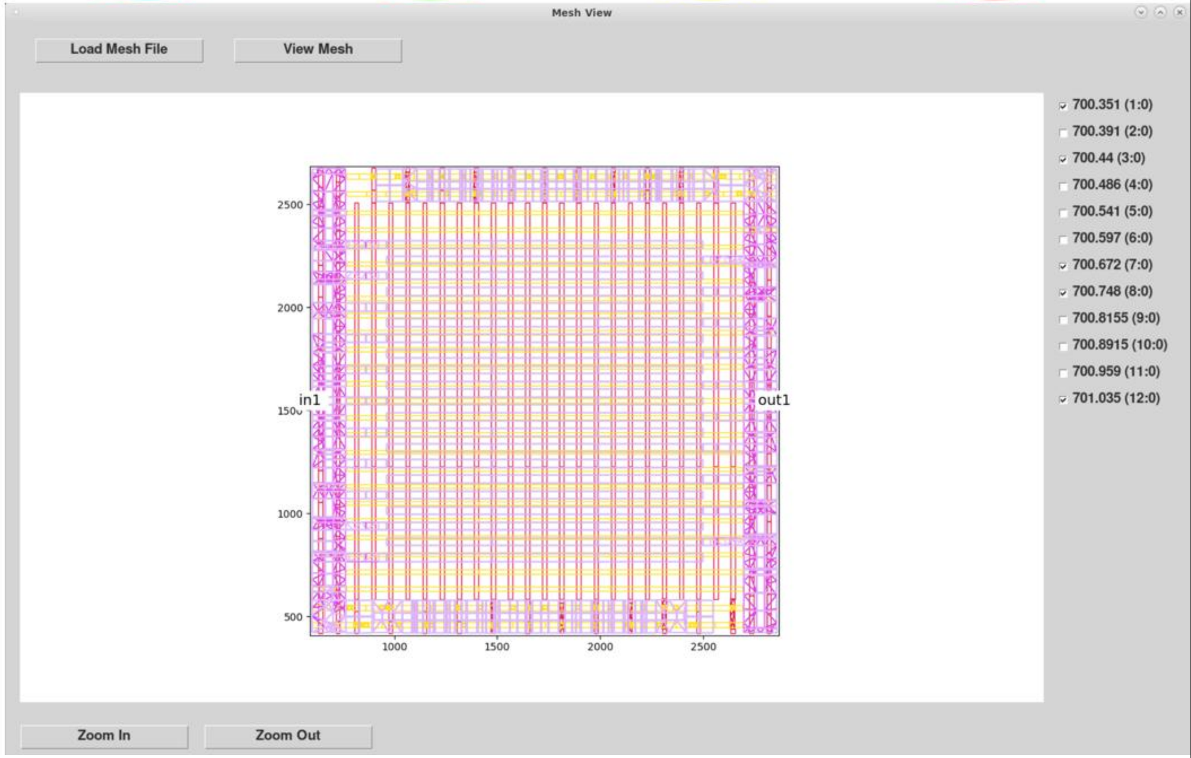
Improved



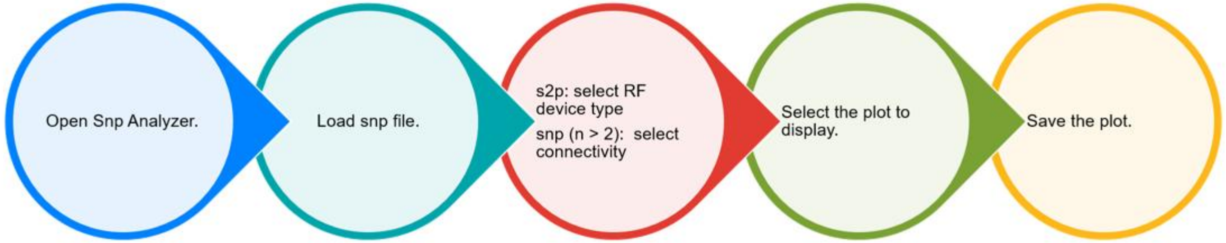


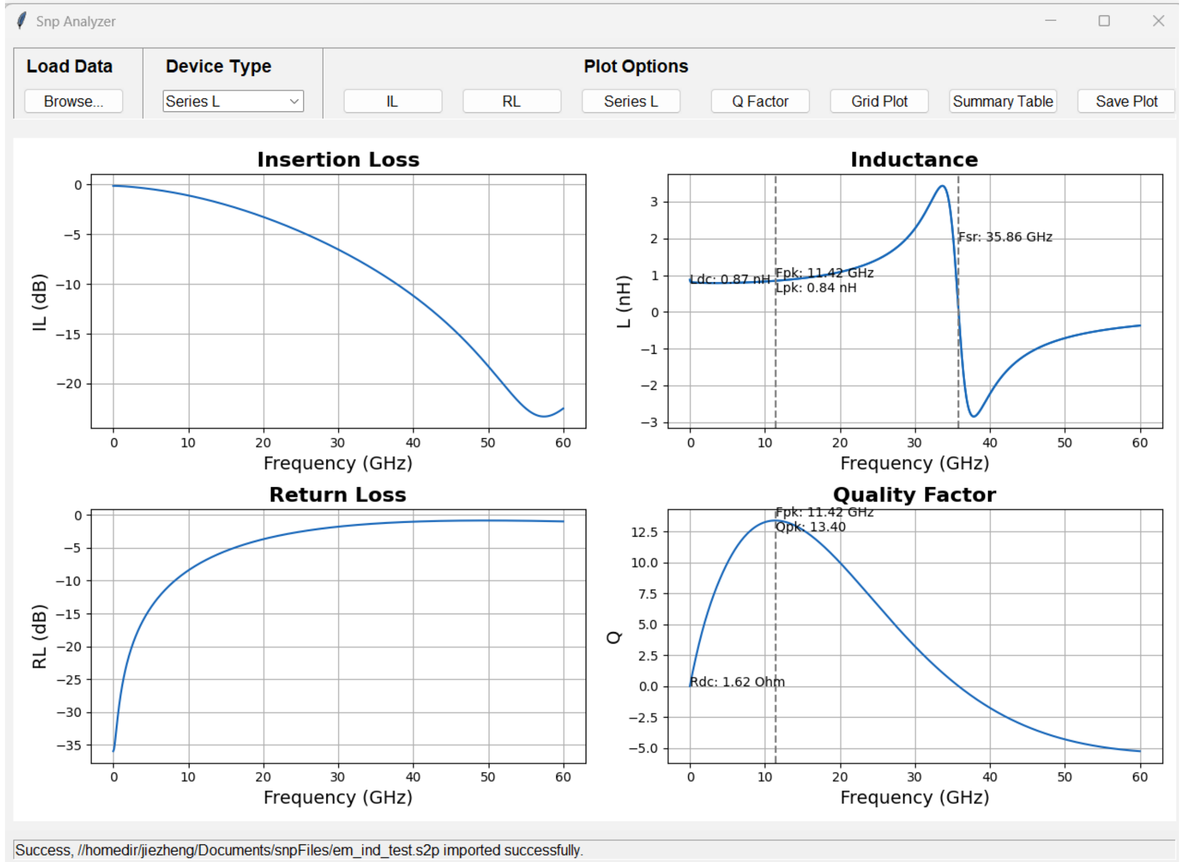
Mesh View



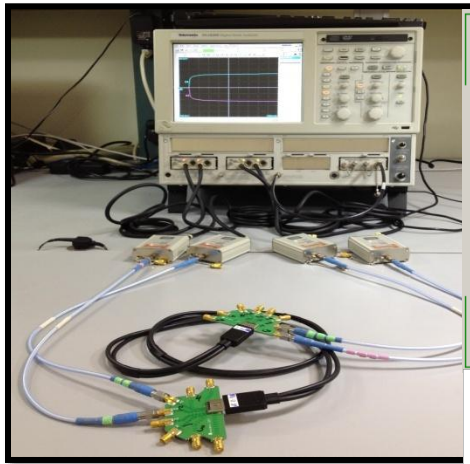


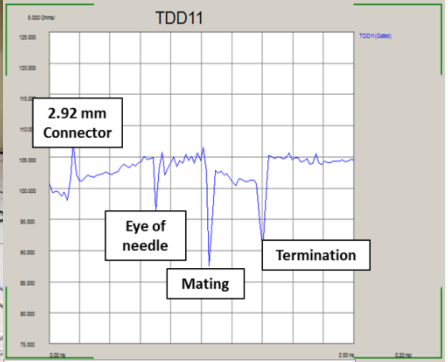
Snp View

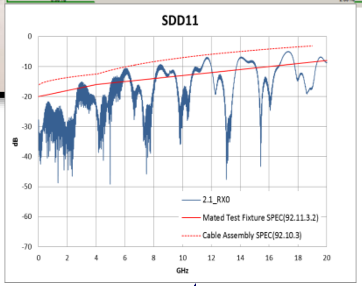




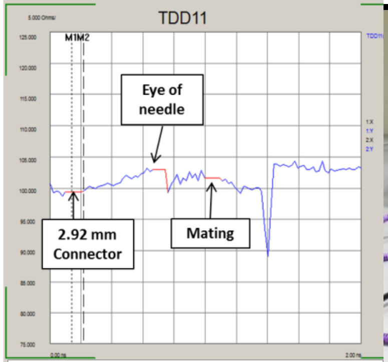
Time-Domain Measurement

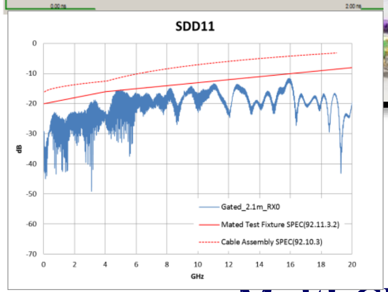


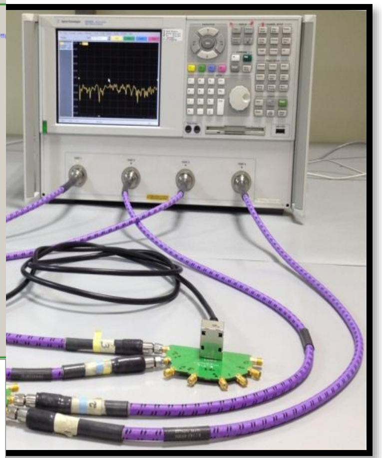




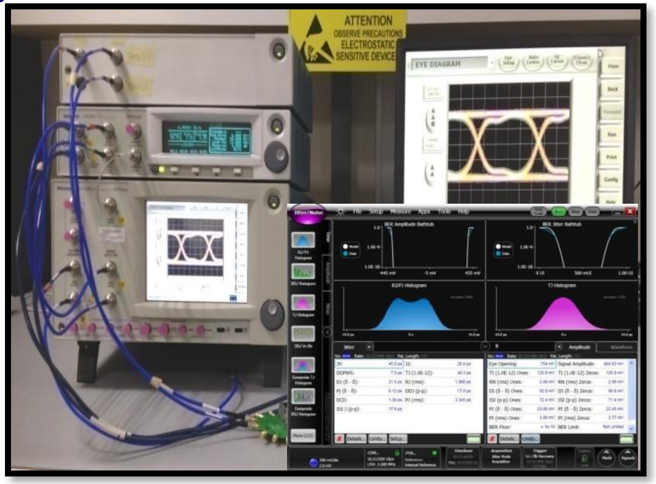
Frequency-Domain Measurement



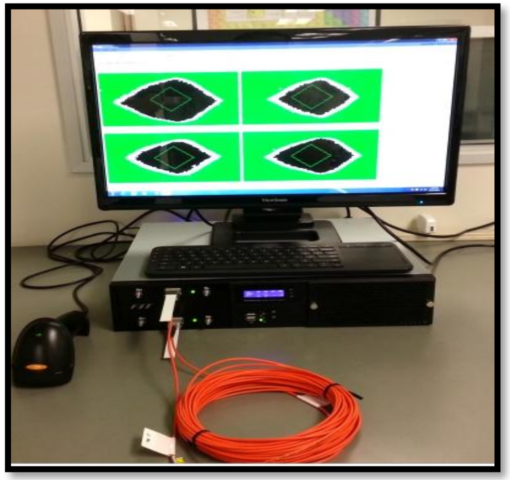




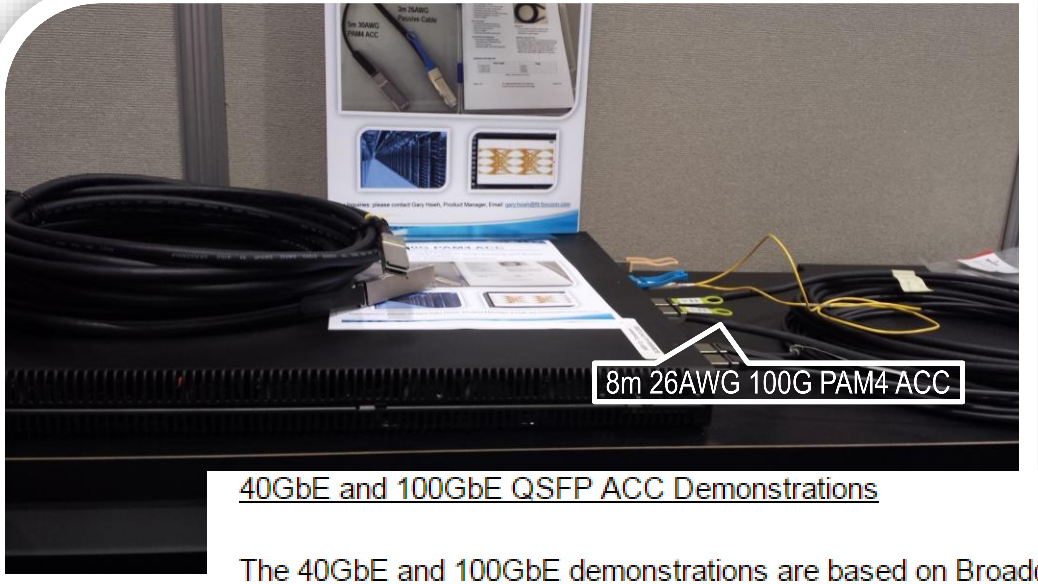
Eye-Pattern/Jitter/BER Measurements



Multi-Channel Auto-Tester



2016 OFC Demonstration of 100G PAM4 ACC

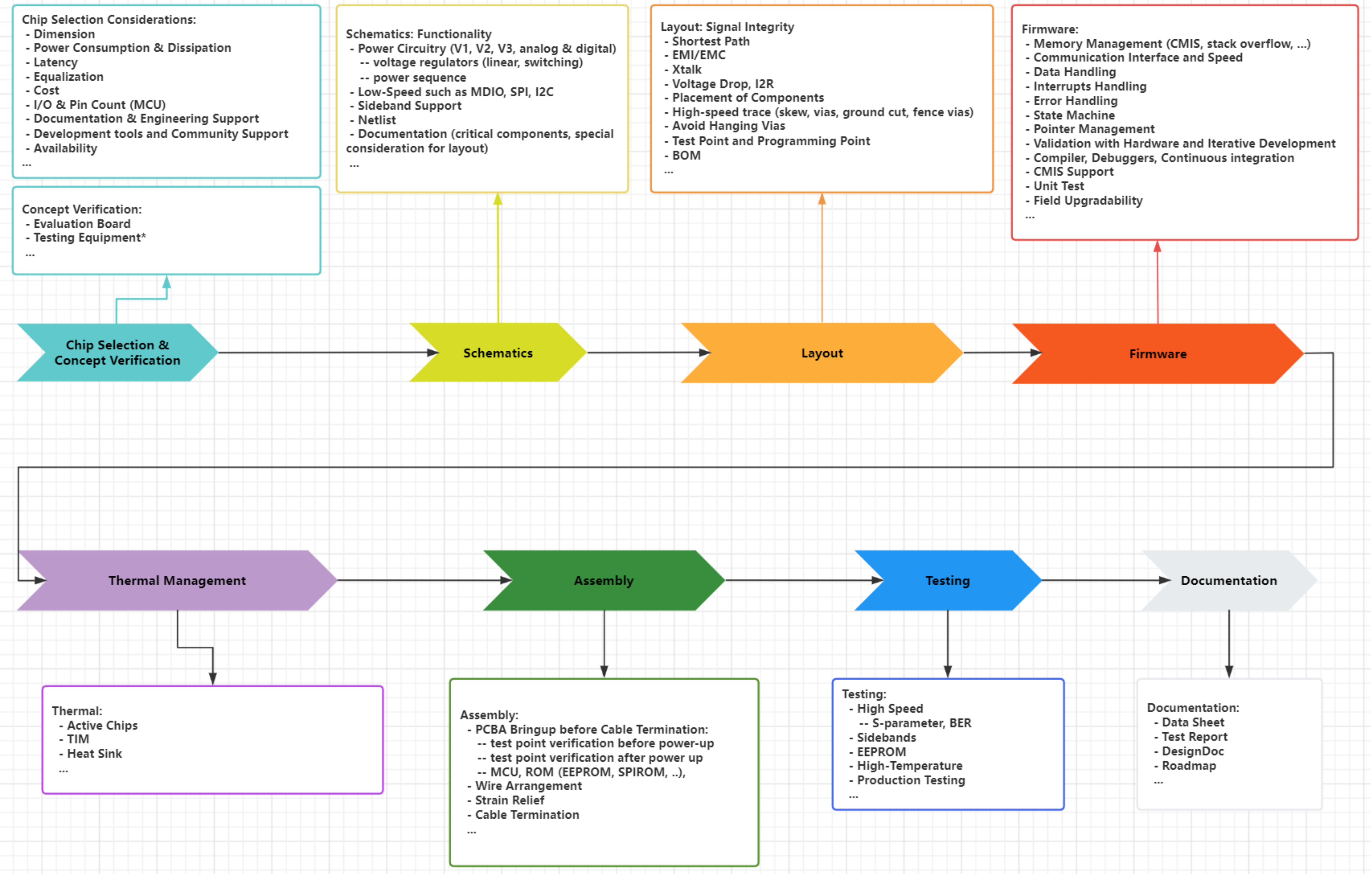


40GbE and 100GbE QSFP ACC Demonstrations

The 40GbE and 100GbE demonstrations are based on Broadcom's low power, low latency PAM-4 PHY chips with integrated retimer and equalizer functions, the BCM59240 and BCM59025, and the latest thin active copper cable (ACC) solutions from Foxconn Interconnect Technology. In the 40GbE demonstration, attendees will see live traffic stream of 40GbE data transmitted up to 8 meters of 26AWG QSFP ACC. In the 100GbE demonstration, attendees will see live traffic stream of 100GbE data transmitted up to 7 meters of 26AWG QSFP ACC.

"We are pleased to have collaborated with Broadcom in demonstrating 40GbE and 100GbE data center interconnects over extended copper cables. The combination of FIT's latest QSFP ACC solution and Broadcom's PAM-4 PHY provides a lower cost alternative to conventional active optical cable (AOC) solutions for short reach 40GbE and 100GbE applications," said Yan Margulis, vice president of sales and marketing at Foxconn Interconnect Technology.

http://www.stockhouse.com/news/press-releases/2016/03/16/broadcom-demonstrates-industry-leading-pam-4-phy-chips-supporting-40-100-200gbe


Chip Selection Considerations:
- Dimension
- Power Consumption & Dissipation
- Latency
- Equalization
- Cost
- I/O & Pin Count (MCU)
- Documentation & Engineering Support
- Development tools and Community Support
- Availability
...
Concept Verification:
- Evaluation Board
- Testing Equipment*
...
Schematics: Functionality
- Power Circuitry (V1, V2, V3, analog & digital)
-- voltage regulators (linear, switching)
-- power sequence
- Low-Speed such as MDIO, SPI, I2C
- Sideband Support
- Netlist
- Documentation (critical components, special consideration for layout)
...
Layout: Signal Integrity
- Shortest Path
- EMI/EMC
- Xtalk
- Voltage Drop, I2R
- Placement of Components
- High-speed trace (skew, vias, ground cut, fence vias)
- Avoid Hanging Vias
- Test Point and Programming Point
- BOM
...
Firmware:
- Memory Management (CMIS, stack overflow, ...)
- Communication Interface and Speed
- Data Handling
- Interrupts Handling
- Error Handling
- State Machine
- Pointer Management
- Validation with Hardware and Iterative Development
- Compiler, Debuggers, Continuous integration
- CMIS Support
- Unit Test
- Field Upgradability
...
Chip Selection & Concept Verification
Schematics
Layout
Firmware
Thermal Management
Assembly
Testing
Documentation
Thermal:
- Active Chips
- TIM
- Heat Sink
...
Assembly:
- PCBA Bringup before Cable Termination:
-- test point verification before power-up
-- test point verification after power up
-- MCU, ROM (EEPROM, SPIROM, ..),
- Wire Arrangement
- Strain Relief
- Cable Termination
...
Testing:
- High Speed
-- S-parameter, BER
- Sidebands
- EEPROM
- High-Temperature
- Production Testing
...
Documentation:
- Data Sheet
- Test Report
- DesignDoc
- Roadmap
...

Foxconn Interconnect Technology Showcases OSFP 2x1 SMT Connector Upper Port with Synopsys 224G Ethernet PHY at DesignCon 2024

March 13, 2024 0 Comments

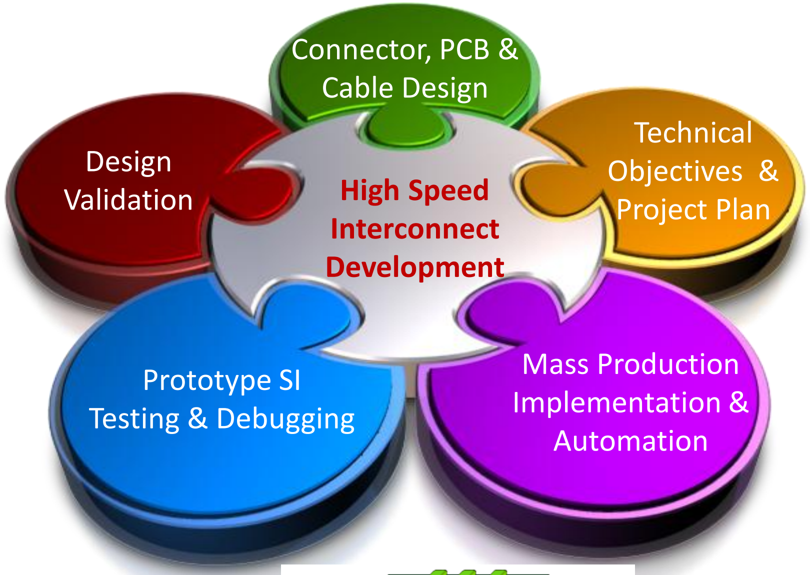
KEYWORDS 224 GBPS DESIGNCON FIT OSFP 2X1 SMT SYNOPSYS

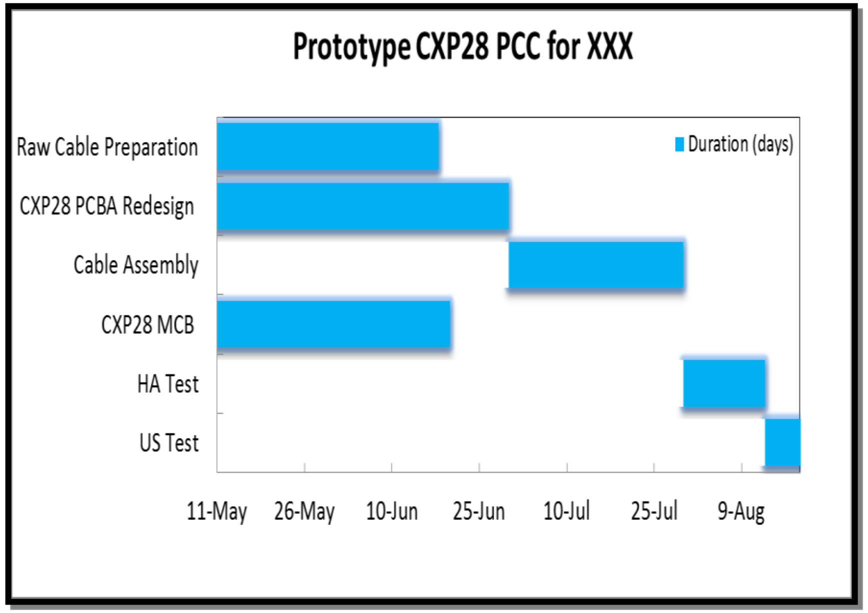
Foxconn Interconnect Technology (FIT), a leading connective solution provider creating powerful connections for a better world, showcased the demonstration of its 224 Gbps signal transmission through their OSFP 2x1 SMT connector Upper Port with a 1M Passive DAC at the DesignCon 2024 exhibition in Santa Clara, Calif. from January 30 to February 1.

FIT's OSFP 2x1 SMT connector, cage, and Module Compliance Board (MCB) appeared together with Synopsys silicon-proven 224G Ethernet PHY IP. This demo featured the Synopsys 224G PHY IP, connected with FIT's OSFP 2x1 Module Compliance Board mounted with the OSFP SMT 2x1 connector and cage. It linked with a 1M passive DAC cable, to another OSFP 2x1 MCB on the other end. This upper port SMT connector link operating at 224 Gbps is very attractive in the industry, as the SMT connector lead-frames in the upper port are very long and challenging to accommodate the stringent signal integrity requirements. This demo focused on the Signal Integrity requirements of the upcoming IEEE 802.3dj specification which is still in progress. The connector and cage are also backwards compatible to all other variants of OSFP Cables and Connectors.

https://www.signalintegrityjournal.com/articles/3508-foxconn-interconnect-technology-showcases-osfp-2x1-smt-connector-upper-port-with-synopsys-224g-ethernet-phy-at-designcon-2024

- Define the real goal
- Bound the problem
- Trade-offs explicitly
- Mechanism over heroics
- Risk control





HS SI Fixture

Design & Analysis for SI
-PCB material & SMA/2.92mm
-Anti-pad and stack-up for Z & XT
-Trace length for IL, Rise Time

Calibration Design
-SOLT
-TRL (>3GHz)
-De-embedding (AFR, ISD)

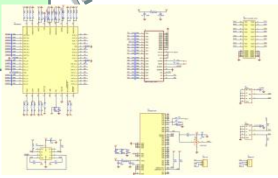
Layout

Function Board

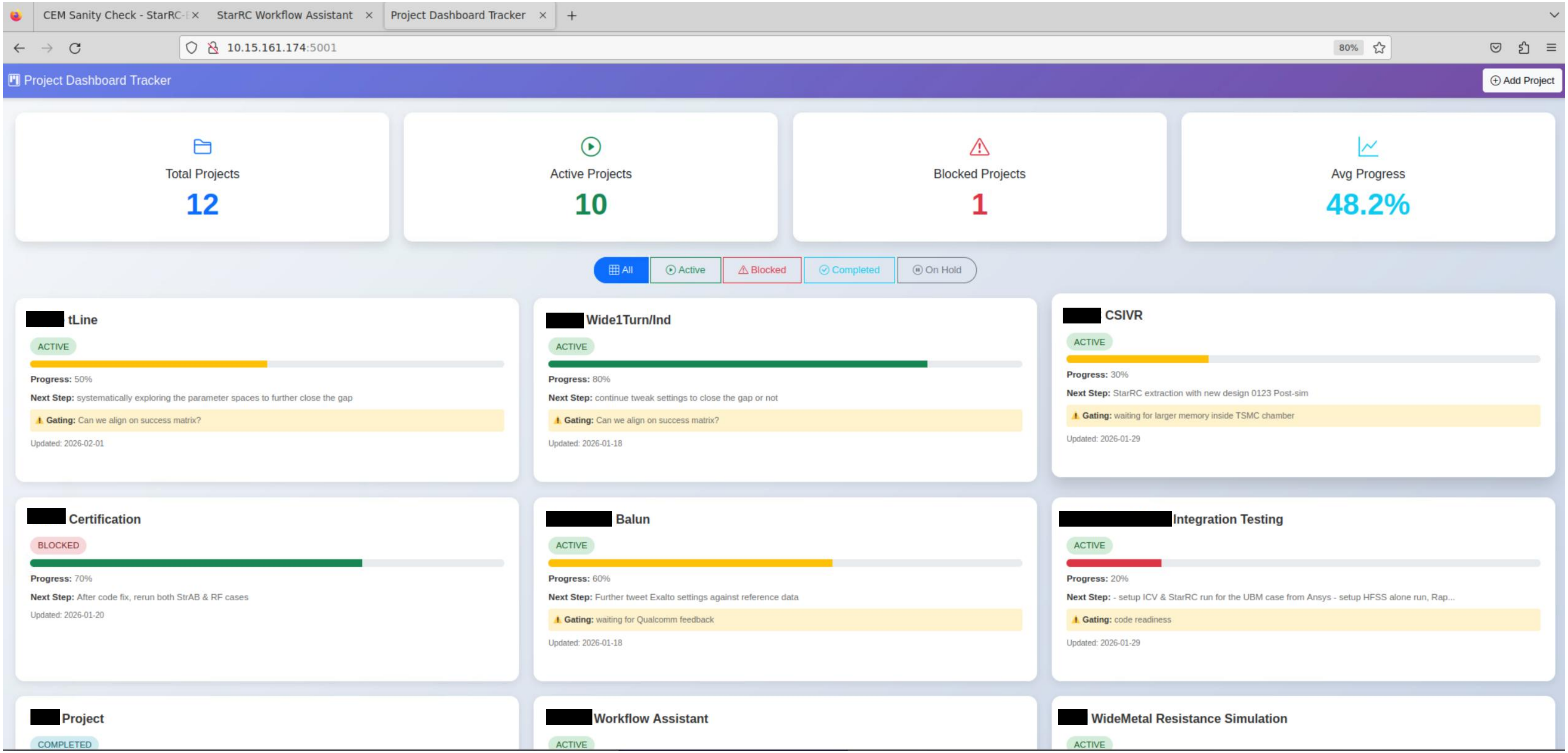
System Design

Pre-Layout Analysis

Circuit Design & Schematic



Transparency Across Team



Data Center / Enterprise

- IEEE 802.3dj, 802.3cd, 802.3bj, 802.3ba
- Ethernet 10G, 25G, 50G, 100G
- PCIe Gen I, II, III, IV
- SAS-1, SAS-2, SAS-3, SAS-4
- Fibre Chanel 8G, 10G, 16G, 32G/128G
- InfiniBand SDR, DDR, QDR, FDR, EDR, HDR, NDR

Consumer

- USB 1.0, 2.0, 3.0, 3.1 Gen1 & Gen2, 4.0
- HDMI 1.0, 1.1, 1.2, 1.3, 1.4, 2.0, 2.1
- DisplayPort 1.1, 1.2, 1.3, 1.4, 2.0, 2.1
- Thunderbolt v1, v2, v3

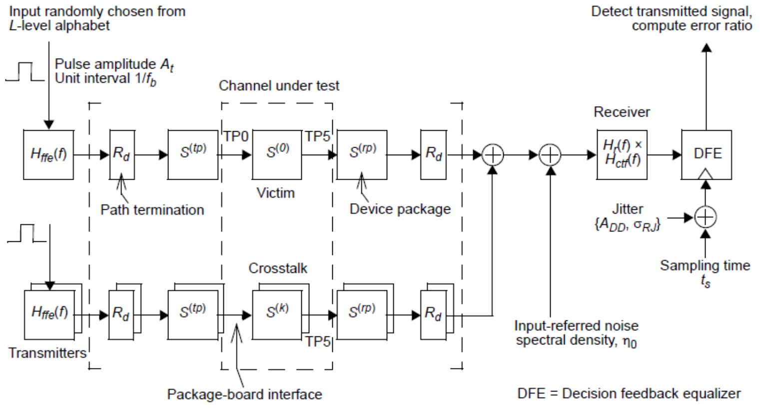
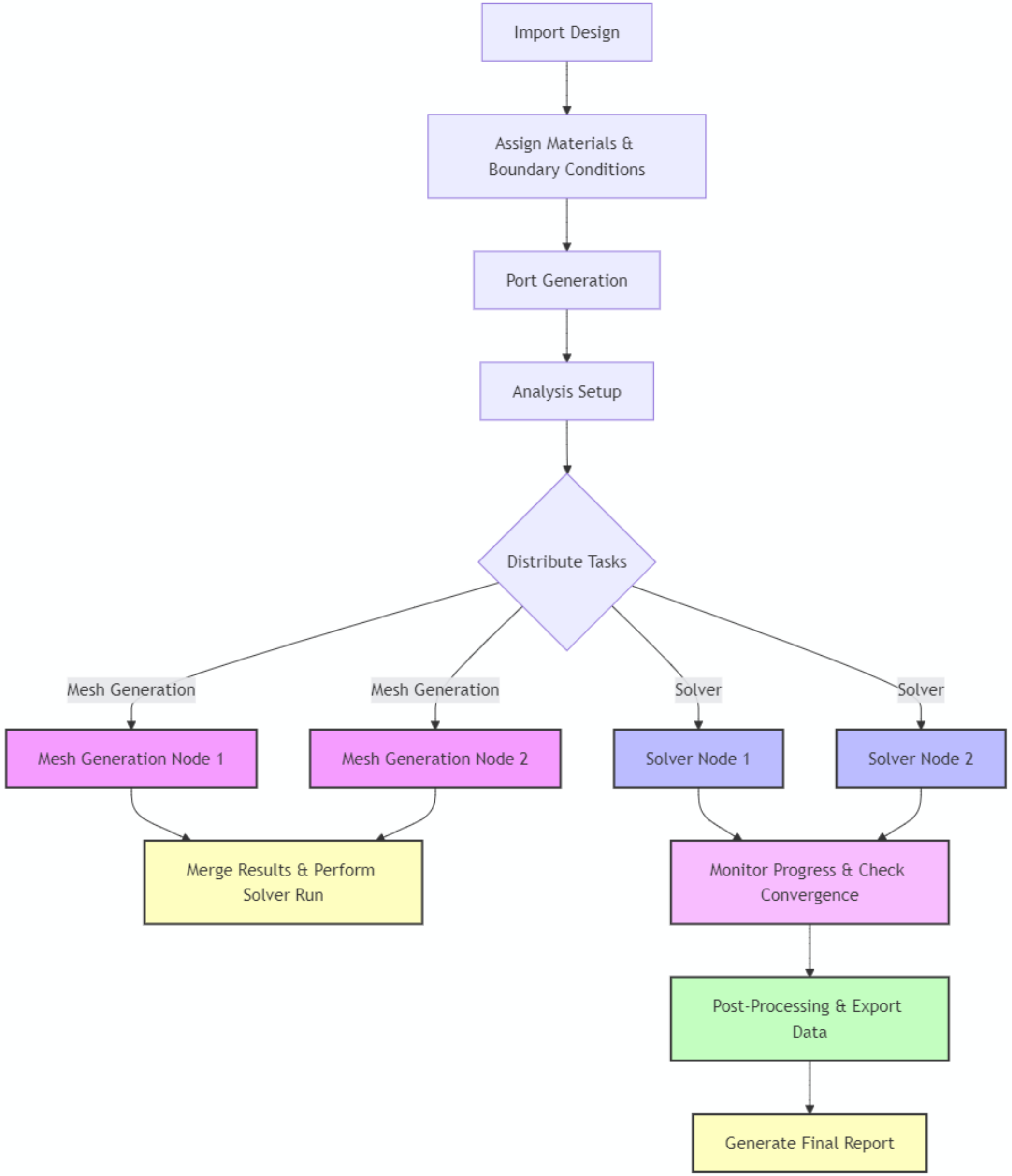


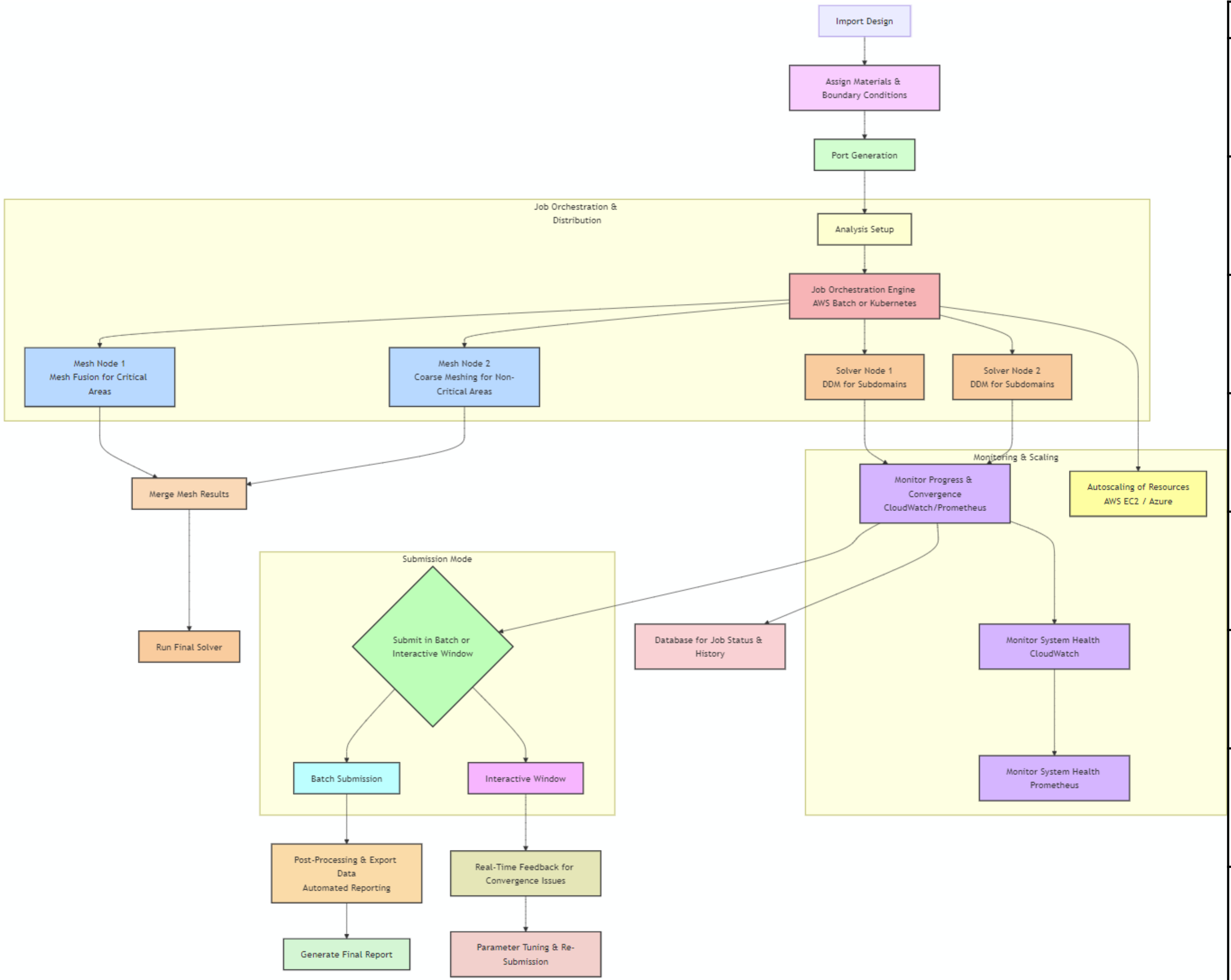
Figure 93A–1—COM reference model




```python
from pyaedt import Hfss

hfss = Hfss()

# Import
hfss.import_gds_3d("path_to_model.gds")

# assign material
hfss.assign_material( assignment: "GroundPlane",  material: "Copper")

# Create_wave_port(location=[0, 0, 0], port_name="Port1")
hfss.wave_port( assignment: "M2",  reference: "G2", location=[0, 0, 10], port_name="Port2")
# Create a lumped port
hfss.lumped_port( assignment: "GroundPlane", location=[5, 5, 0], name="LumpedPort1")

# Define perfect electric conductor (PEC) boundary
hfss.create_boundary( boundary_type: "MyBoundary",  assignment: "PEC", faces=[1, 2, 3])

# Create a setup with frequency sweep for S-parameter analysis
setup = hfss.create_setup(setupname="Setup1")
setup.props["Frequency"] = "2.5GHz"
setup.props["MaxPasses"] = 20

# Set up a frequency sweep
hfss.create_linear_count_sweep(
    setupname="Setup1",
    sweepname="Sweep1",
    startfreq="1GHz",
    stopfreq="3GHz",
    count=100
)


# Run the simulation and monitor its progress
hfss.solve_in_batch(filename="my_run_file") # run_solution("Setup1")
```

https://aedt.docs.pyansys.com/version/0.9/API/_autosummary/pyaedt.hfss.Hfss.html#

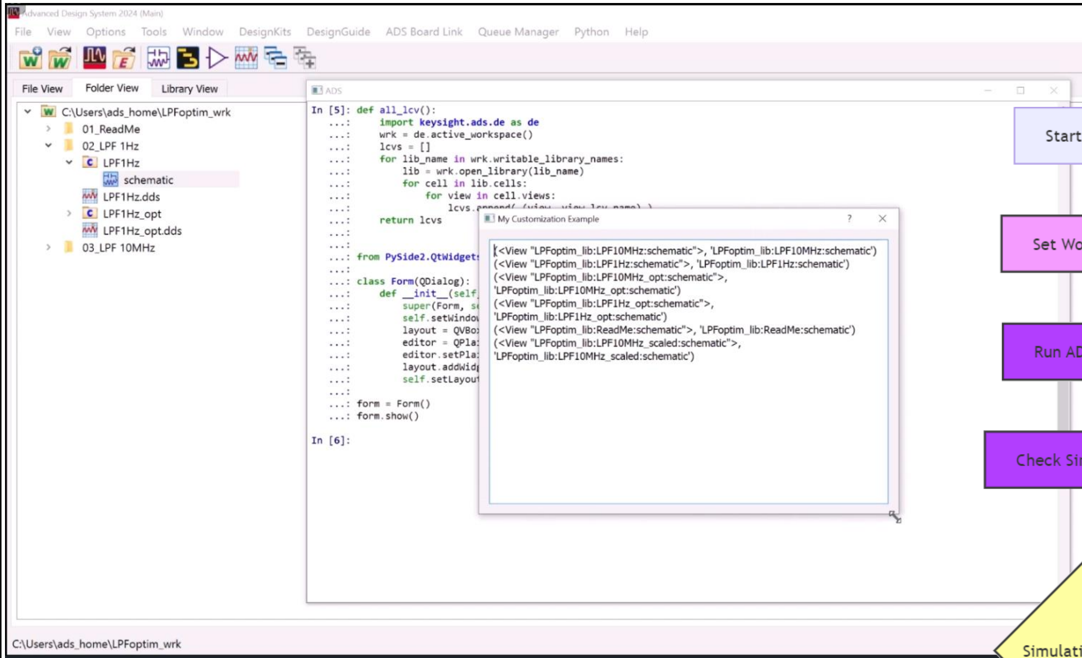
Design Automation Flow



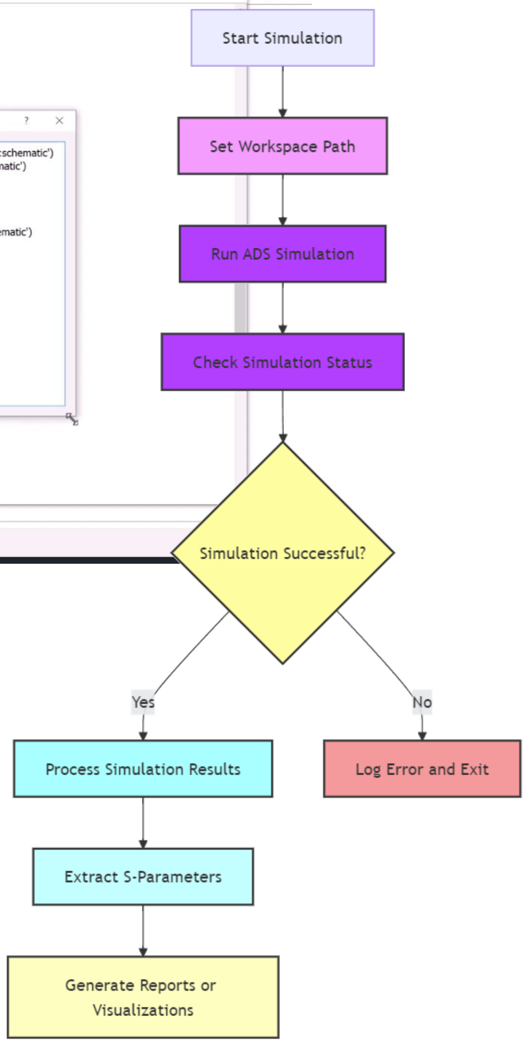
Category	Implementation	Management
Domain Decomposition Method	Develop scripts to split MCM designs into **subdomains** for parallel simulations using DDM.	Oversee **resource allocation** across cloud environments (AWS EC2/Azure), ensuring optimal cost ratios.
Mesh Fusion for Optimized Meshing	Use **Mesh Fusion** to apply high-detail meshing for critical areas (antennas) and coarse meshing for others.	Guide teams on how to **balance accuracy vs. performance**, setting clear priorities for critical areas.
Cloud HPC Resource Allocation	Automate **job orchestration** using AWS Batch/Kubernetes to assign resources dynamically.	Ensure **cloud compute resources** are allocated cost-effectively and manage infrastructure budgets.
Monitoring and Health Checks	Implement **CloudWatch** or **Prometheus** for real-time **monitoring** and health checks for simulations.	Monitor **resource usage** across nodes, ensuring failures are detected early and rescaling is automatic.
Submission Modes	Enable **batch submission** for automated simulations or **interactive mode** for real-time parameter tuning.	Manage **team-wide workflows** to decide between batch or interactive submissions based on project needs.
Post-Processing and Reporting	Automate **post-processing** and report generation using Python scripts or AWS Lambda.	Ensure **timely reporting**, managing the expectations of teams and optimizing the post-processing pipeline.
Pre-Layout & Post-Layout Feature	Allow fast initial design and iteration run	
Tacit Knowledge Integration	Use PEC for symmetrical structures, …	

```python
from skillbridge import Workspace

# Step 1: Open Virtuoso Workspace
ws = Workspace.open()

# Step 2: Get the current cell view (MCM design)
cv = ws.ge.get_edit_cell_view()

# Step 3: Fetch all instances in the design (e.g., components like capacitors, tr
instances = cv.instances()

# Step 4: Print instance statistics
print(f"Total number of instances: {len(instances)}")
for inst in instances:
    print(f"Instance Name: {inst.name}, Type: {inst.type}, Position: {inst.xy}")

# Step 5: Modify an instance (e.g., change the position of an instance)
if instances:
    inst_to_modify = instances[0]  # Modify the first instance
    print(f"Original Position: {inst_to_modify.xy}")
    inst_to_modify.move([100, 100])  # Move instance to a new position
    print(f"New Position: {inst_to_modify.xy}")

# Step 6: Save the modified design
cv.save()
print("Design saved successfully.")
```

https://unihd-cag.github.io/skillbridge/
https://github.com/unihd-cag/skillbridge




```python
# Step 3: Process the simulation results (e.g., S-parameters)
def process_simulation_results(result_file):
    # Open the results file and extract relevant data
    with open(result_file, "r") as f:
        for line in f:
            freq, s11, s21 = line.split()
            print(f"Frequency: {freq} GHz, S11: {s11} dB, S21: {s21} dB")


# Step 4: Run the workflow
design_name = "mcm_design"
run_ads_simulation(design_name)

# Step 5: Specify the location of the results file and process it
result_file = "/path/to/sim_results/s_parameters.txt"
process_simulation_results(result_file)
```



https://www.keysight.com/us/en/assets/3123-1748/demos/ADS-Python-Automation-for-Developers.mp4?courseId=345&nextLessonNumber=9

Design / Layout

- **Define Success Matrix**
  - IL/RL/NEXT/FEXT/Sdc
  - EQ assumptions (CTLE/DFE/FFE/IBIS-AMI)

- **Stack up**:
  - low-loss laminate
  - copper surface roughness
  - tighten outer-layer signal to adjacent reference plane
  - surface finish (ENIG/ENEPIG/ImAg…)

- **Routing discipline**:
  - controlled impedance
  - symmetry + skew
  - short parallelism
  - increase space

- **Return path continuity**
  - no plane splits
  - stitch near transitions
  - controlled reference transitions

- **Discontinuities engineered**:
  - vias/back drill/anti-pad

Validate / Correlate

- **Fixture strategy first**:
  - IEEE P370
  - On-board standards/coupons

- **Simulation:**
  - HFSS, HFSS 3D Layout

- **TDR/TDT**:
  - locate discontinuities;
  - verify impedance profile
  - skew

- **VNA**:
  - Sdd21/Sdd11/Sdc21
  - NEXT/FEXT
  - scan for resonance

- **Robustness**:
  - tolerance/DOE on critical parameters
  - screening strategy
  - production control plan

Design

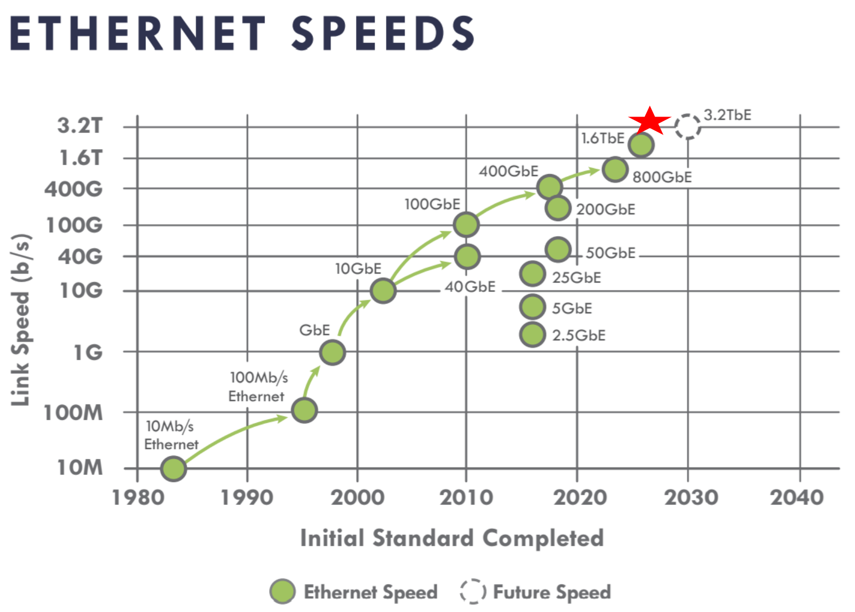
- **Define Success Matrix**
  - IL/RL/NEXT/FEXT/Sdc/COM
  - EQ assumptions (CTLE/DFE/FFE/IBIS-AMI)
- **Return path topology**:
  - continuous ground engagement
  - cage grounding
  - shield strategy
- **Control asymmetry**:
  - minimize differential ↔ common conversion through symmetry
- **Crosstalk strategy**:
  - containment (grounds/walls/shields)
  - spacing
  - shorten coupling length
  - uniformity
- **Resonance Suppression**:
  - avoid cavities/periodic pinfields;
  - damp high-Q resonances

Proof / Validation

- **Multi-domain Co-Simulation**
  - EM (HFSS)
  - circuit (ADS)
  - Channel simulation with IBIS-AMI
- **Multiport VNA plan**:
  - mixed-mode S-params;
  - de-embedding
- **BER**:
  - eye diagram
  - BER
- **Time-domain fingerprints**:
  - TDR signature windows for discontinuities
  - root-cause localization
- **Variation → production**:
  - manufacture tolerance
  - plastic Dk drift
  - Mating repeatability

	SIMULATION Side	MEASUREMENT Side
Physics / Model Issues	**Q1 – Model Deficiency**	**Q3 – Setup Error**
	Geometry not true-to-fab	Fixture or reference plane
	Material model	Calibration
	Missing physics	Port Mapping
	Incomplete solver settings	Incorrect post-processing
Manufacturing / Assembly Variations	**Q2 – As-Built Deviation**	**Q4 – DUT Variation**
	Process drift	Sample-to-sample spread
	Tolerance stack-up	Mating repeatability
	Assembly Damage	Environmental conditions

Feature	200GbE	400GbE	800GbE	1.6 Terabit	3.2 Terabit
Standard	IEEE 802.3bs/cd	IEEE 802.3bs/ck	IEEE 802.3df	IEEE 802.3dj (Draft)	Research OIF
Primary Lane Rate	50 Gbps	50 / 100 Gbps	100 Gbps	200 Gbps	400 Gbps (Target)
Lane Count	4x 50G	8x 50G 4x 100G	8x 100G	8x 200G	8x 400G
Modulation	PAM4	PAM4	PAM4	PAM4	PAM4 / PAM6+?
Symbol Rate	26.56 GBd	26.56 / 53.1 GBd	53.125 GBd	106.25 GBd	212+ GBd
Key Enabler	Transition from NRZ	DSP maturity	100G SerDes	200G SerDes	Co-Packaged Optics (CPO)

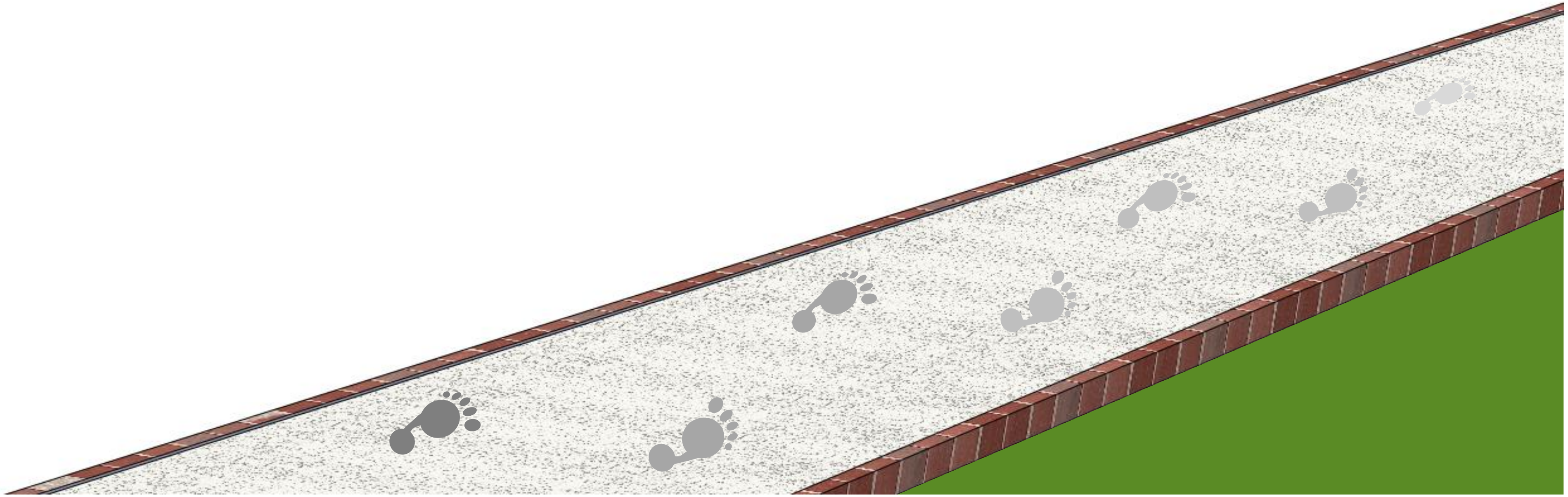


https://ethernetalliance.org/wp-content/uploads/2025/12/EthernetRoadmap-2026-Side2-Press-3.pdf

- **Mission**:
  - Creating a Safer, Sustainable, Productive, and Connected Future
- **Recent News**:
  - TE Connectivity launches 56G MezzaWave connectors and cable assemblies
  - Advancing end-to-end optical infrastructure for next-generation AI data centers at OFC 2026
  - TE Connectivity acquires EV charging inlet assets from Phoenix Contact
  - ATOP and TE Connectivity Deepen Collaboration on 6.4T CPO Ahead of OFC 2026
- **Scale**:
  - Presence in 130 countries
  - 90,000+ Employees
  - 150,000+ Patents
  - 235 Billion Products Manufactured Annually
- **Culture**:
  - Integrity
  - Accountability
  - Inclusion
  - Innovation
  - Teamwork

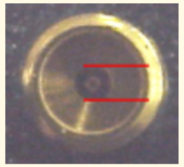
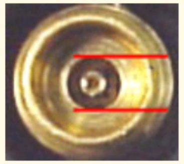
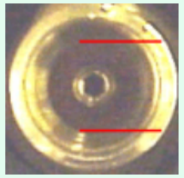
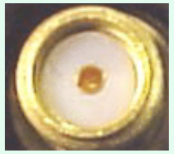
https://www.te.com/en/home.html
https://careers.te.com/

Thank You !

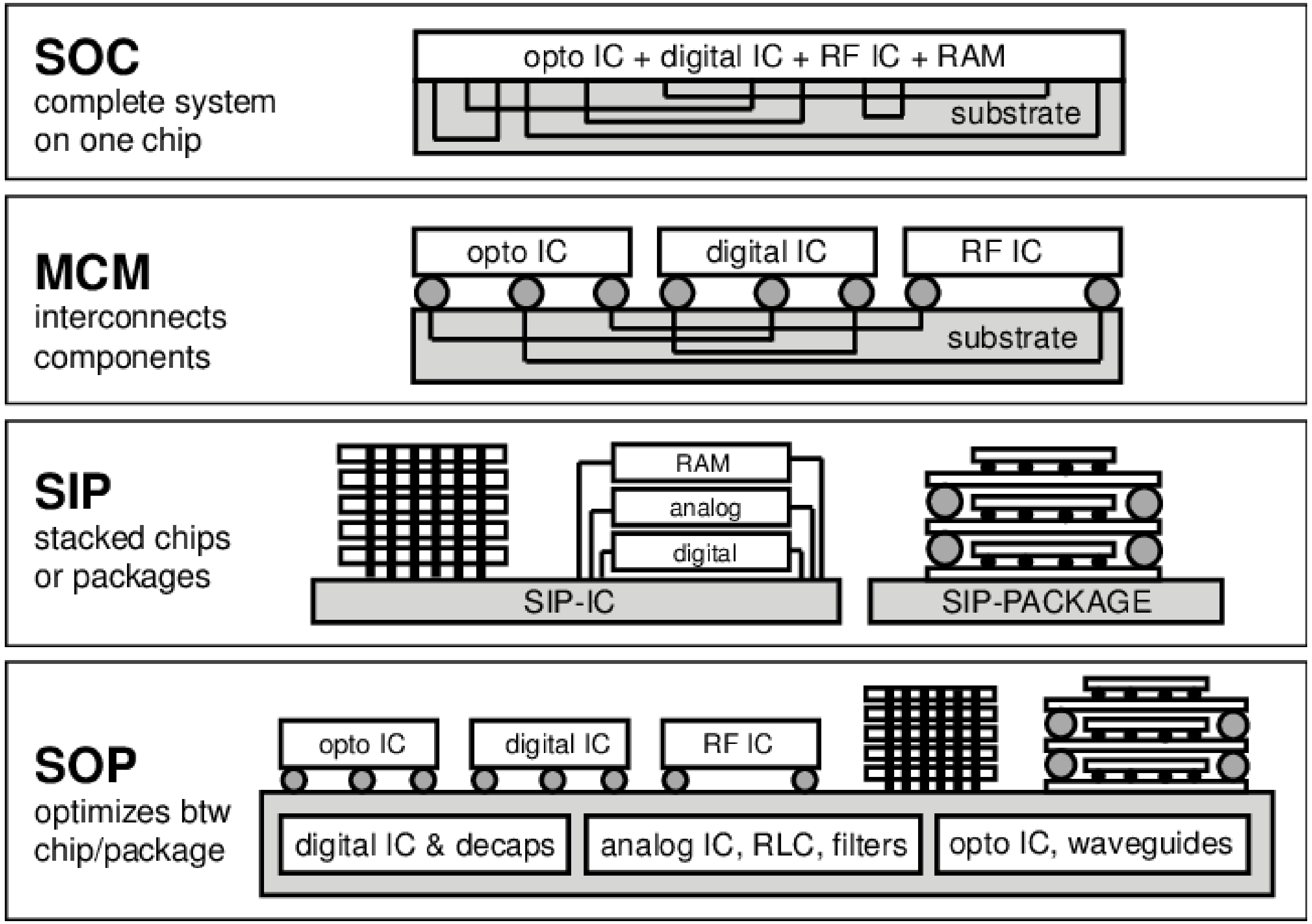


Backup Slides

RF Connectors

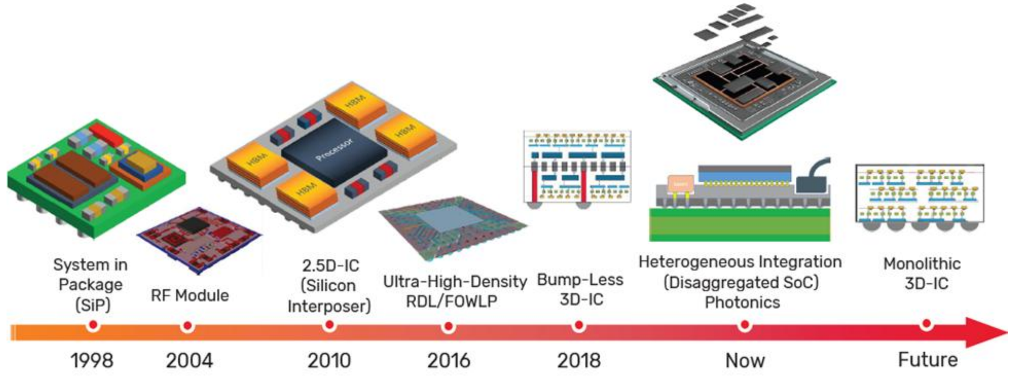
Connector Type	Frequency Range	Mates with...	Notes
 1.0 mm	To 110 GHz	1.0 mm	Much smaller connector than any of those below.
 1.85 mm	To 70 GHz	2.4 mm	The outer thread size of the 1.85 and 2.4 connectors is bigger than SMA, 3.5, and 2.92. This makes the area of the outer conductor mating surface look very large compared to the relatively small air dielectric.
 2.4 mm	To 50 GHz	1.85 mm	The 1.85 mm connector that is manufactured at Keysight has a groove in the male nut and female shoulder to distinguish these two connector types.
 2.92 mm	To 40 GHz	3.5mm and SMA	The 3.5 mm and 2.92 mm connectors use the same center pin.
 3.5 mm	To 34 GHz	2.92 mm and SMA	The 3.5 mm and 2.92 mm connectors use the same center pin.
 SMA	To 24 GHz	2.92 mm and 3.5 mm	Uses a PTFE dielectric.

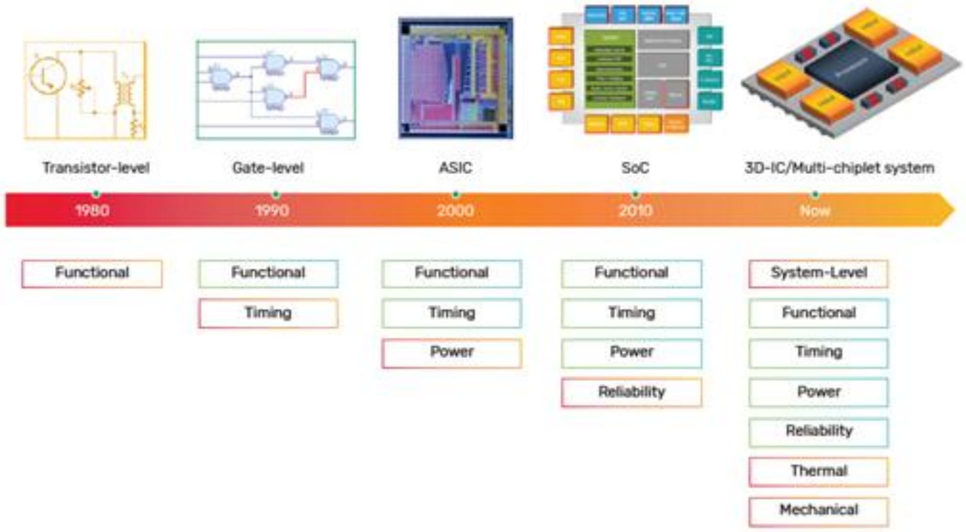
https://docs.keysight.com/kkbopen/what-rf-connectors-are-mateable-749648236.html



Comparison among SOC (System-On-Chip), MCM (Multi-Chip Module), SIP (System-In-Package), and SOP (System-On-Package).

https://www.researchgate.net/figure/Comparison-among-SOC-System-On-Chip-MCM-Multi-Chip-Module-SIP-System-In-Package_fig1_228870734





https://community.cadence.com/cadence_blogs_8/b/di/posts/adopting-a-faster-more-efficient-path-to-multi-chiplet-design